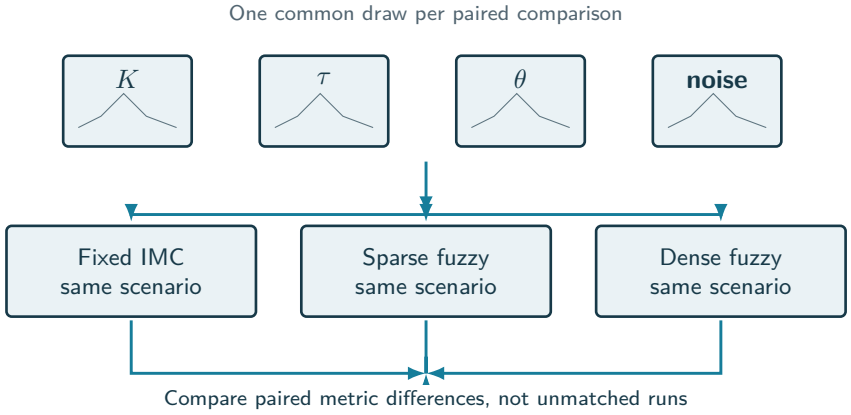
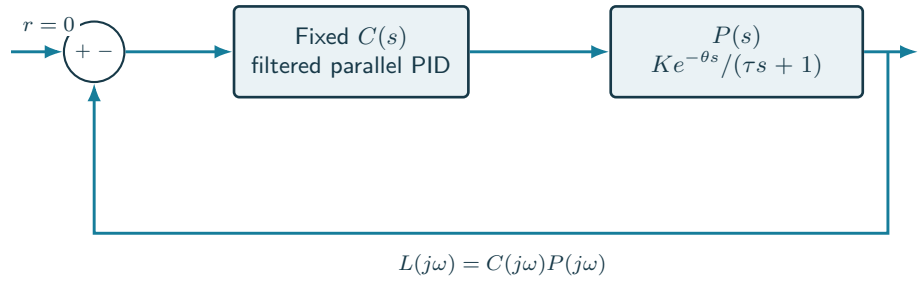
Robustness: paired uncertainty and a separate linear loop audit

A. Nonlinear / sampled comparison
Freeze a controller design, draw synthetic plant parameters, then run every controller against the same parameter draw, noise seed, limits and disturbance schedule. Pairing reduces comparison noise; it does not validate the chosen probability distributions.



B. Fixed, unsaturated return-ratio model
Analyze one fixed PID and one FOPDT plant, with the reference held at zero for the loop test. Report the frequency grid and crossover method. Scheduling and clipping are omitted; margins do not certify the nonlinear fuzzy loop.



Report a vector of consequences
Rise time and overshoot before disturbances; full-horizon IAE; command total variation; saturation fraction; post-closure recovery with a sustained tolerance band. Lower error can require more actuator movement. Record both.

Sensitivity and margins

$S(j\omega) = 1/[1 + L(j\omega)], \quad M_s = \max_{\omega \in \Omega} |S(j\omega)|.$

Gain and phase margins use the first detected crossings on an explicit grid. No crossing on that grid is not proof of an infinite physical margin. Inspect additional crossings and Nyquist behavior when needed.

Missing physical evidence
A single thermal state does not predict sensor-to-sensor uniformity, component temperature, heater calibration or equipment interlocks. Synthetic parameter sweeps and linear-loop margins address declared numerical models only.

The distributions drawn above are conceptual symbols, not fitted chamber distributions. Numerical uncertainty and frequency-response plots remain generated by the notebooks and robustness.py, with their original seeds and model assumptions.
Original explanatory drawing for ThermoFuzzy-PID. Source traceability: Cordeiro (UFC, 2021), documented repository equations and explicit reconstruction choices.